

LESSON 13

LINEAR PROGRAMMING

Overview:

In this lesson, students will develop techniques needed to solve system of equalities and graph systems of equalities. They will extend these skills to solve real-worlds scenarios, applicable through production problems (minimize cost and/or maximize profit) when given various constraints.

Learning Outcomes:

At the end of this lesson, the students can:

1. graph the feasible region;
2. represents constraints by equation or inequalities, and by system of equations and/or inequalities;
3. solve linear programming problems; and
4. apply linear programming in solving real life situation problems.

Materials Needed:

- Modules (hard and soft copy)
- Devices (laptop and mobile phone)
- Social Media Platforms (Moodle, Google classroom, Facebook, YouTube and etc.).
- Graphing paper
- Pencil/ballpen
- Ruler

Duration: 3 hours

Learning Contents:

13.1 The Linear Programming

Linear Programming is a mathematical technique used to find the optimal allocation of resources in the military, business, telecommunications, and other fields.

To solve linear programming problems, we must maximize (or minimize) a function (called the **objective function**) subject to the given restrictions on its variables. These restrictions (called **constraints**) are usually given as a system of linear inequalities.

EXAMPLE Suppose that the annual profit (in millions of pesos) earned by a business is given by the equation $P = 2x + y$ and that x and y are subject to the following constraints:

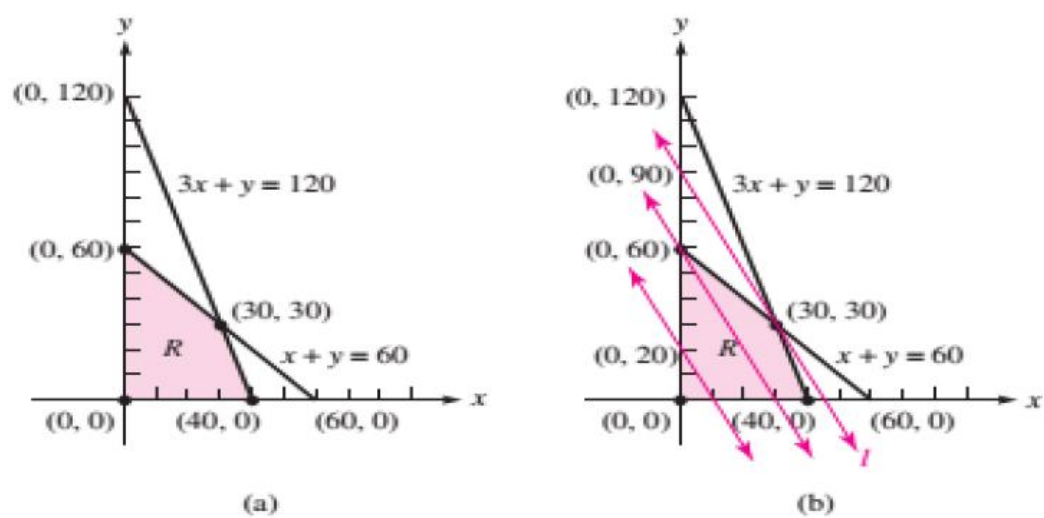
$$\begin{cases} 3x + y \leq 120 \\ x + y \leq 60 \\ x \geq 0 \text{ and } y \geq 0 \end{cases}$$

Solution:
To find the maximum profit P that can be earned by business, we solve the system of inequalities as shown in the figure 13(a) and find the coordinates of each corner point of the region . This region is often called **feasibility region**. We can then write the profit equation:

$$P = 2x + y \text{ in the form of } y = -2x + P$$

The equation is the equation of a set parallel lines, each with a slope of -2 and a y y-intercept of P . The graph of for these values of P is show in red lines in Figure 13(b). to find the red line that passes through region R and provides the maximum value of P , we locate the red line with the greatest y-intercept. Since the line has the greatest y-intercept and intersect region at the corner point $(30, 30)$, the maximum value of (subject to the given constraints) is

Thus, the maximum profit that can be earned is Php. 90 million. This profit occurs when .



13.2 MAXIMUM OR MINIMUM OF AN OBJECTIVE FUNCTION

If a linear function, subject to the constraints of a system of linear inequalities in two variables, attains a maximum or a minimum value, that value

will occur at a corner point or along at the entire edge of the region that represent the solution of the system.

EXAMPLE: If $P = 2x + 3y$, find the maximum value of P subject to the following constraints:

$$\begin{cases} x + y \leq 4 \\ 2x + y \leq 6 \\ x \geq 0, y \geq 0 \end{cases}$$

Solution:

We solve the system of inequalities to find the feasibility region shown in figure 13.1. The coordinates of each corner pints are $(0, 0)$, $(3, 0)$, $(0, 4)$, and $(2, 2)$.

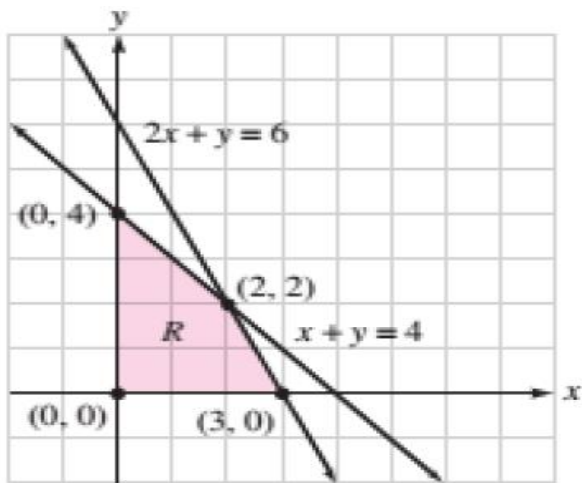


Figure 13.1

Thus, the maximum value occurs when

Since the maximum value of will occur at a corner of , we substitute the coordinates of each corner point into the object function and find the one that gives the maximum value.

Point	$P = 2x + 3y$
$(0, 0)$	$P = 2(0) + 3(0) = 0$
$(3, 0)$	$P = 2(3) + 3(0) = 6$
$(2, 2)$	$P = 2(2) + 3(2) = 10$
$(0, 4)$	$P = 2(0) + 3(4) = 12$

Strategy for Solving Linear Programming Problems

1. Find the objective function and constraints.
2. Find the feasibility region by graphing the system of inequalities and identifying the coordinates of its corner points.
3. Find the maximum (or minimum) value by substituting the coordinates of the corner points into the objective function.

EXAMPLE

An accountant prepares tax returns for individual and for small businesses. On average, each individual requires 3 hours of her time and 1 hour of computer time. Each business return requires 4 hours of her time and 2 hours of computer time. Because of other business considerations, her time is limited to 240 hours, and the computer is limited to 100 hours. If she earns a profit of Php. 80. 00 on each individual return and a profit of Php. 150. 00

on each business return, how returns of each type should she prepare to maximize her profit.

Solution:

Let us make a table for the given information.

	Individual tax return	Business tax return	Time available
Accountant's time	3	4	240 hours
Computer time	1	2	100 hours
Profit	P80	P150	

Find the objective function and constraints.

Let x represents the number of individual tax returns and y for the number of business tax returns to be completed. Take note that each of the x individual tax return will earn Php. 80.00 profit while each y business tax return will earn Php. 150. 00. Hence, the total profit is given by the equation

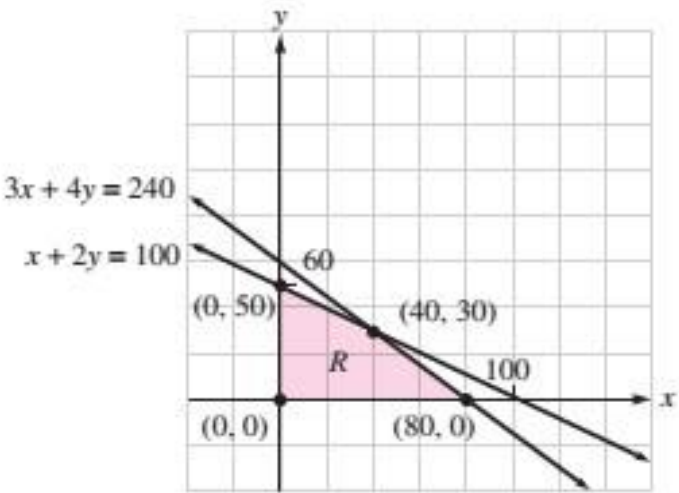
$$P = 80x + 15y \text{ (Objective Function)}$$

We know that the number of each return, either individual or business, cannot be negative. Thus, $x \geq 0$ and $y \geq 0$.

Every x individual tax return takes 3 hours of the accountant's time and each y business tax return needs 4 hours of her time, that is $(3x + 4y)$ hours she will be working. This amount must be less than or equal to her 240 hours available time. Thus, the inequality $3x + 4y \geq 240$ is a constraint to the accountant's time.

Moreover, since every individual tax return takes 1 hour of computer time and each business tax return needs 2 hours of its time, that is $(x + 2y)$ hours the computer will be working. This amount must be less than or equal to the computer available time of 100 hours. Thus, the inequality $x + 2y \geq 100$ is a constraint to the accountant's time.

Constraints:



Point	
(0, 0)	
(80, 0)	
(40, 30)	
(0, 50)	

From the table, we can see that the accountant will earn a maximum profit of ₱7,700 if she prepares 40 individual returns and 30 business returns.

Learning Activity:

Group Activity. Write the objective function and the inequalities that describe the constraints in each problem. Graph the feasibility region, showing corner points. Then find the maximum or minimum value of the objective function.

Group 1: Two woodworkers, Tom and Carlos, get ₱1000 for making a table and ₱800 for making a chair. On average, Tom must work 3 hours and Carlos 2 hours to make a chair. Tom must work 2 hours and Carlos 6 hours to make a table. If neither wishes to work more than 42 hours per week, how many tables and how many chairs should they make each week to maximize their income? Find the maximum income.

Group 2: Two artists, Nina and Rob, make yard ornaments. They get ₱80 for each wooden snowman they make and ₱64 for each wooden Santa Claus. On average, Nina must work 4 hours and Rob 2 hours to make a snowman. Nina must work 3 hours and Rob 4 hours to make a Santa Claus. If neither wishes to work more than 20 hours per week, how many of each ornament should they make each week to maximize their income? Find the maximum income.

Group 3: An electronics store manager stocks from 20 to 30 IBM-compatible computers and from 30 to 50 Apple computers. There is room in the store to stock up to 60 computers. The manager receives a commission of ₱50 on the sale of each IBM-compatible computer and ₱40 on the sale of each Apple computer. If the manager can sell all of the computers, how many should she stock to maximize her commissions? Find the maximum commission.

Group 4: A diet requires at least 16 units of vitamin C and at least 34 units of vitamin B complex. Two food supplements are available that provide these nutrients in the amounts and costs shown in the table. How much of each should be used to minimize the cost?

Learning Evaluation:

Solve the following problems.

- 1) A high school band trip will require renting buses and trucks to transport no fewer than 100 students and 18 or more large instruments. Each bus can accommodate 40 students plus three large instruments; it costs ₱350 to rent. Each truck can accommodate 10 students plus 6 large instruments and costs ₱200 to rent. How many of each type of vehicle should be rented for the cost to be minimum? Find the minimum cost.
- 2) Does the objective function attain a maximum at the corners of a region defined by following nonlinear inequalities? Attempt to maximize on the region and write a paragraph on your findings.
- 3) Minimize the objective function of the previous problem.
- 4) Maximize P subject to the following constraints.

References:

- Aufman, R. et. al. (2018). *Mathematics in the Modern World*. Rex Book Store Inc. Sampaloc, Manila, Philippines.
- Polya, G. (1957). *How to Solve It*. 2nd Edition. Princeton University Press.